Q.1 The Simplex method is a---------- for solving Linear Programming problem.

A. computational procedure(Algorithm)

B. graphical procedure

C. analytical procedure

D. probabilistic procedure

Q.2 While solving a Linear Programming problem by Simplex Method, an inequality says:'2x-3y>5' is transformed into strict equality of the form---------, where 's>=0'

A. 2x-3y –s =5

B. 2x-3y –s +A =5

C. 2x-3y +s + A=5

D. 2x-3y +s =5

Q.3 When we have \geq type constraints, then we convert it to equality constraints by introducing _____ variable(s) and solve it by _____ method.

A. slack, Big M

B. artificial, Two phase

C. artificial and slack, Big M or Two phase

D. artificial, Big M

Q.4 In two phase method if the minimum value of objective function in the first phase is zero, then the solution of original problem _____.

A. non of these

B. is uncertain

C. does not exists

D. exists

Q.5 After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables. If the initial basic feasible solution is say;(2,4,1,2), then its corresponding non-basic solution is---------.

A. (-4,-2)

B. (2,2)

C. (1,2)

D. (0,0)

Q.6 By Simplex method, to minimize 'Z = 9x–2y'of an LP problem, if 'z=A>0' for the initial iteration then for its next improved solution(0<A<100), which of the following would be the next entering variable?

A. x>0

B. y<0

C. x<0

D. y>0

Q.7 The inequality 3x - 2y \geq 14 is equivalent to_____.

A. 3x - 2y 14

B. 3x - 2y < - 14

C. - 3x + 2y > 14

D. - 3x + 2y - 14

Q.8 In the Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column's entries should be --------- except key row(pivot) entry.

A. strictly positive

B. strictly negative

C. zero

D. maximum or minimum

Q.9 By Simplex method, to maximize 'Z = 2x–9y'of an LP problem, if 'z=0' for the initial iteration then for its next improved solution, which of the following would be the next entering variable?

A. y>0

B. y<0

C. x>0

D. x<0

Q.10 While solving an LP problem by Simplex method, the inclusion of slacks in the constraints' inequalities helps in finding ---------- variables.

A. basic

B. non-basic

C. artificial

D. decision
